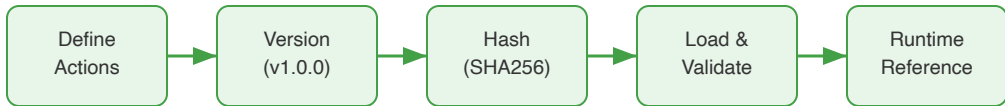


Figure 2: Registry Lifecycle — Version, Hash, Validate, Evolve



Evolution: append-only (safe) → reorder (safe) → incompatible rename (KVRM: 0% stale, classifier: 19-44%)